

Scheduled, Simultaneous Dosing of Pregabalin, Celecoxib, and Acetaminophen Markedly Reduces or Eliminates Opioid Use After ACL Reconstruction Using Allograft or Hamstring Tendon Autograft

A Randomized Clinical Trial

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Background: Opioid analgesics continue to be prescribed after ambulatory surgery despite untoward adverse effects, risk of overdose, and association with substance use disorder.

Purpose/Hypothesis: The purpose was to investigate the use of a novel system to provide scheduled and simultaneous dosing of acetaminophen, celecoxib, and pregabalin after anterior cruciate ligament reconstruction (ACLR). It was hypothesized that this system would markedly reduce pain and opioid use compared with existing best practice.

Study Design: Randomized controlled trial; Level of evidence, 1.

Methods: Included were 100 patients scheduled for elective, primary ACLR using allograft or hamstring tendon autograft. Selection criteria included age between 18 and 65 years and weight between 65 and 120 kg. Exclusion criteria were a known allergy to any drug used in the study or the use of opioid analgesics before surgery. Patients in the intervention group received a blister pack with scheduled, simultaneous doses of acetaminophen, celecoxib, and pregabalin; patients were also given oxycodone 5 mg as needed for breakthrough pain. Patients in the control group were prescribed ibuprofen and oxycodone 5 mg/acetaminophen 325 mg as needed for pain. The primary outcome measure was pain. Secondary outcomes were nausea, itching, and daily oxycodone use. Patients were asked to quantify their average pain at rest, nausea, and itching on an 11-point verbal scale (from 0 to 10). These data were recorded for 6 days during daily telephone contacts with patients after hospital discharge.

Results: Cumulative results for 6 days showed significantly lower values in the intervention group compared with the control group for pain (median [interquartile range], 28 [14-35] vs 35 [28-41], respectively; $P = .009$) and oxycodone use (median [interquartile range] number of tablets, 0 [0-2] vs 8 [1.25-16], respectively; $P < .001$). Based on these data, the upper tolerance limits for the number of oxycodone tablets required by 90% of patients in the intervention and control groups were 8 tablets and 30 tablets, respectively. Cumulative results for nausea and itching were also significantly lower for the intervention group. Most patients in the intervention group used no opioids during recovery.

Conclusion: Simultaneous dosing of 3 nonopioid analgesics resulted in reduced postoperative pain and markedly lower opioid use.

Registration: NCT04015908 (ClinicalTrials.gov identifier).

Keywords: postoperative pain; opioids; multimodal; analgesia

reluctant to withhold opioids in the setting of moderate to severe postoperative pain. To this end, a variety of procedural and pharmacologic treatments have been used to reduce opioid requirements after hospital discharge.

Many existing protocols for reduced or “opioid-free” recovery simply impede or eliminate access to opioid analgesics.^{5,12,15,17,23} This practice may reduce opioid use at the expense of inadequately treating pain. Truly effective nonopioid analgesia should decrease opioid use even for patients with unrestricted access to opioids during recovery.

Three classes of nonopioid analgesics are practical for use in an outpatient setting. Acetaminophen, nonsteroidal anti-inflammatory drugs (NSAIDs), and gabapentinoids all produce significant reductions in opioid requirements and opioid-related side effects.^{16,21,24} To date, there are limited data describing the simultaneous effect of all 3 drug classes after hospital discharge.¹⁷ Patient compliance with complex drug regimens may limit the use of multimodal analgesia after hospital discharge.²²

In this prospective randomized trial, we investigated the use of a novel drug packaging system to facilitate the utilization of 3 commonly prescribed nonopioid analgesics after ambulatory orthopaedic surgery (anterior cruciate ligament reconstruction [ACLR]). We hypothesized that improved compliance with the scheduled, simultaneous administration of these 3 nonopioid analgesics would markedly reduce pain and opioid use after primary ACLR compared with existing methods for pain control.

METHODS

This study received institutional review board approval and was registered with ClinicalTrials.gov (NCT04015908). Written informed consent was obtained from all participants. Consecutive eligible patients having elective, isolated, and primary ACLR using either semitendinosus autograft or allograft between January 2020 and February 2021 were asked to participate in the study. Enrollment was limited to patients having hamstring autograft or allograft in order to minimize variability of surgical stimulus that may be associated with other graft options, such as patellar or quadriceps tendon. Additional criteria for inclusion were age between 18 and 65 years and weight between 65 and 120 kg. Patients were excluded if they had a known allergy to any drug used in the study, were using opioid analgesics before surgery, or did not consent for adductor canal block (ACB) for postoperative pain management.

There was no masking of study group assignment for surgeon, anesthesiologist, research personnel, or patient. Intention-to-treat analysis was used for all participants.

A total of 167 consecutive patients were evaluated for enrollment. From this total, 65 patients did not meet inclusion criteria, and 2 patients (1 male and 1 female) declined to participate; thus, 100 patients were ultimately enrolled (Figure 1). After agreeing to enroll, patients were randomized to either the intervention or control group ($n = 50$ patients each), with concealment of the allocation process. The randomization schedule was generated in blocks of 10 numbers (Research Randomizer; <http://www.randomizer.org>). One patient in the control group was lost to follow-up.

Participants received 1 of 2 regimens for pain management for a period of 6 days after hospital discharge. Patients in the intervention (blister pack) group received a novel blister pack of nonopioid analgesics containing 2 acetaminophen caplets 325 mg, celecoxib 100 mg, and pregabalin 50 mg to be taken as scheduled doses 4 times a day for 6 days (Figure 2). This group was also given oxycodone 5 mg (40 tablets) to be taken every 4 hours (1-2 tablets) as needed for pain not controlled by the nonopioid analgesics.

Patients in the control group were instructed to use ibuprofen up to 800 mg every 8 hours as needed for pain control. In addition, patients received combined oxycodone 5 mg/acetaminophen 325 mg (40 tablets) to be taken as needed for pain every 4 hours (1-2 tablets). Patients in the control group were also allowed to use plain acetaminophen and instructed not to exceed 650 mg every 6 hours.

One hour before surgery, all participants received standardized oral multimodal analgesics consisting of acetaminophen 975 mg, celecoxib 400 mg, and pregabalin 150 mg. All patients with a known history of postoperative nausea and vomiting or motion sickness were treated 1 hour before surgery with a transdermal scopolamine patch (1.5 mg) that was left in place for 72 hours after application. Patients in both groups were provided with equipment for cold-pad knee circulation therapy at home after discharge from the postanesthesia care unit (PACU).

Anesthesia was induced with intravenous propofol 1.5 mg/kg and maintained using propofol 0.1 mg/kg/min and remifentanyl 0.2 μ g/kg/min. Intravenous fentanyl was given at the discretion of the anesthesiologist. Both groups received intravenous ondansetron 4 mg and dexamethasone 0.1 mg/kg for nausea prophylaxis.

In the PACU, both groups received an ultrasound-guided ACB performed at the level of the distal third of the femur using a simplified ACB technique.²⁷ Injectate for the ACB consisted of a 30-mL volume of preservative-free 0.5%

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Ethical approval for this study was obtained from the University of Utah (reference No. 00121505).

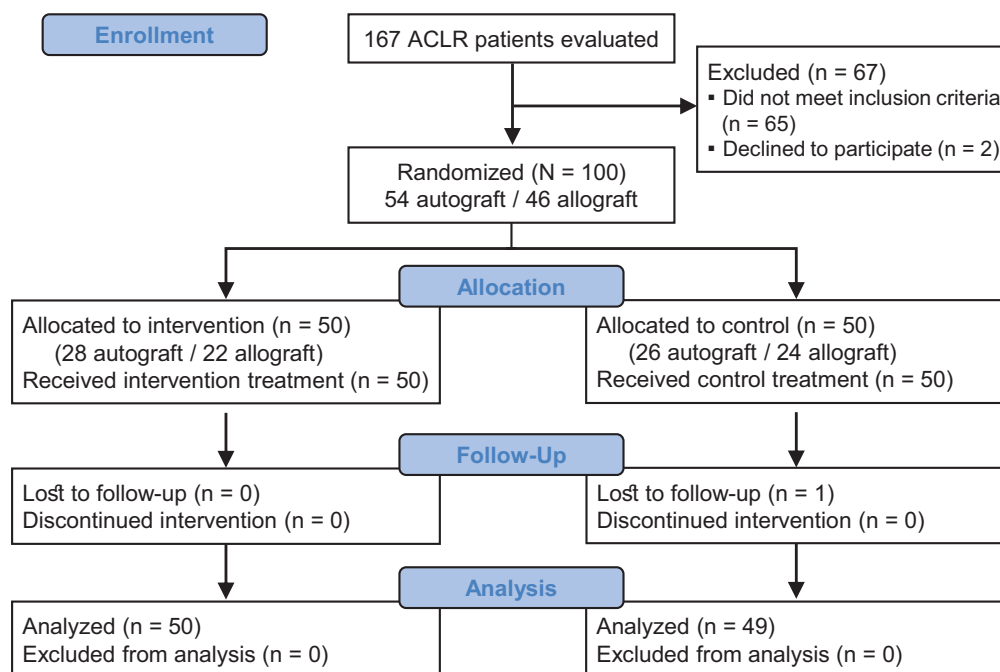


Figure 1. Patient flowchart from enrollment and randomization to allocation, follow-up, and analysis. In total, 65 patients did not meet inclusion criteria because of age or weight ($n = 26$), planned patellar tendon graft ($n = 28$), or recent opioid use ($n = 11$). Note that only 2 patients declined to participate in the trial. A single patient was lost to follow-up after ambulatory surgery. ACLR, anterior cruciate ligament reconstruction.

bupivacaine with epinephrine 5 $\mu\text{g/mL}$. Patients received intravenous fentanyl at the discretion of the PACU nurse to achieve a level of analgesia that was acceptable to the patient while at rest.

Before discharge, patients received pain medications and instructions according to their assigned treatment group. Patients in the intervention group were given a blister pack (Figure 1) for scheduled doses of nonopioid analgesics. In addition, patients receive a prescription for oxycodone 5 mg tablets with instructions to take 1 to 2 by mouth every 4 hours as needed for pain.

Patients in the control group were instructed to take ibuprofen up to 800 mg every 8 hours as needed for pain. In addition, patients received a prescription for combined oxycodone 5 mg/acetaminophen 325 mg, 1 to 2 tablets every 4 hours as needed for pain. Patients were allowed to take plain acetaminophen; however, the total acetaminophen dose was limited to 650 mg every 6 hours. All patients had cell telephone access to the anesthesiologist and the option of an additional ACB within 48 hours of surgery.

A research assistant contacted each patient daily by telephone for 6 days after surgery to document pain scores, opioid requirements, and opioid-related adverse effects. Specifically, patients were asked to quantify their oxycodone tablet consumption, average pain at rest, nausea, and itching on an 11-point verbal scale (from 0 to 10). These scores were recorded during each 12-hour period for 6 consecutive postoperative days.

At the end of the study period, each patient was asked to empty the contents of their opioid medication bottle and

photograph all remaining pills with their cellular telephone. This photograph was sent via a text message to the research assistant to verify opioid use.

Statistical Methods

The sample size was based on an anticipated reduction in opioid consumption of 20% in the intervention group. With a chosen alpha of .05 and setting power of 80%, the sample size per group was 44 patients (<http://ClinCalc.com>). To account for possible attrition of patients, the planned study size was 100 patients with a 1:1 balance of intervention and control groups.

Demographic and anesthetic agent variables were displayed as mean ($\pm\text{SD}$) or count (%). Covariate balance between the intervention and control groups after randomization and before intervention was done by standardized mean difference (SMD). The primary outcome measure was pain reported twice daily for 6 days. The planned secondary outcomes were nausea and itching, also reported twice daily for 6 days. A post hoc secondary outcome was daily oxycodone use (number of tablets) for 6 days. The 11-point verbal scale scores and the count of oxycodone tablets were analyzed as ordinal variables.

Cumulative patient scores for pain, nausea, itching, and tablet use for the 6 days were compared between groups by Wilcoxon rank-sum test with estimation of the median of the difference between intervention and control samples. The 6 days of measurements represent longitudinal data. To avoid assumptions about the probability distribution of

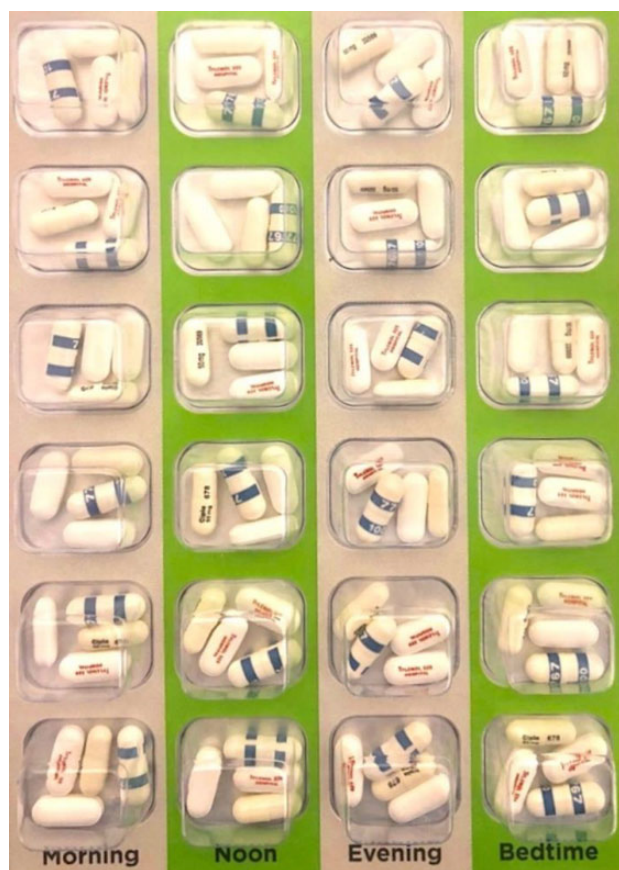


Figure 2. This blister pack drug-delivery system was provided to all patients in the intervention group. There are 6 rows with 4 compartments (morning, noon, evening, bedtime) in each row. Each compartment contains 1 pregabalin 50 mg capsule, 1 celecoxib 100 mg capsule, and 2 acetaminophen 325 mg caplets.

the ordinal outcomes, robust rank-base analysis of variance methods was used to compare intervention and control samples with interaction by postoperative day.²⁰ Also, pairwise comparison for each 12-hour or 24-hour period was estimated by Wilcoxon nonparametric tests, with a false discovery rate adjustment of the *P* values using the Benjamini-Yekutieli procedure.

Tolerance intervals are estimated from a sample and provide limits that at least a certain proportion of a population falls within a given confidence interval.²⁹ In addition to comparing the cumulative opioid requirements for patients in the control and intervention groups, analysis was performed to determine the upper tolerance limit to provide sufficient oxycodone tablets without requiring a refill (the opioid requirements) in 90% or 95% of control and intervention patients, respectively.

All analyses were performed using the R software language (Version 4.1.0) and the R packages nparLD (Version 2.1) and tolerance (Version 2.0.0). No data were imputed for the control patient who was lost to follow-up. The null hypothesis was rejected for $P < .05$; 95% CIs were obtained. Some statistics were rounded.

TABLE 1
Participant Characteristics and Anesthetic Drugs^a

Variable	Blister Pack (n = 50)	Control (n = 50)	SMD ^b
Age, y	32 ± 11	37 ± 11	0.53
Male sex	13 (26)	18 (36)	0.22
Height, cm	169 ± 9	171 ± 8	0.29
Weight, kg	75 ± 13	80 ± 13	0.36
Body mass index	26 ± 6	27 ± 4	0.20
Scopolamine patch	19 (38)	19 (38)	<0.001
Intraoperative fentanyl, µg	110 ± 45	110 ± 46	<0.001
PACU fentanyl, µg	93 ± 66	92 ± 77	0.01

^aData are reported as mean ± SD or n (%). PACU, postanesthesia care unit; SMD, standardized mean difference.

^bThe SMD checks covariate balance after randomization and before the intervention. SMD <0.1 is considered desirable for prognostically important covariates in a trial.

RESULTS

The participant data and anesthetic agents for all 100 participants are shown in Table 1. The distribution of demographic covariates had SMD values exceeding 0.1; the magnitude of differences was not considered clinically relevant. The ratios of semitendinosus autograft/allograft were 28/22 for the intervention and 26/24 for the control groups. There were no additional unanticipated procedures performed in either group.

Three patients in the intervention group and 5 patients in the control group requested and were administered a repeat ACB to reduce pain on postoperative day 2. One patient in the control group refused ACB before discharge from the PACU. Two patients in the intervention group experienced dizziness, and in discussion with the providing physician, this was thought to be possibly related to pregabalin. One of these patients skipped a single dose of pregabalin on postoperative day 3 because of dizziness. The second patient discontinued pregabalin on postoperative day 4 and continued taking only celecoxib and acetaminophen. A third patient in the intervention group reported a facial rash on postoperative day 2, which she attributed to celecoxib. She continued the trial uneventfully by substituting ibuprofen 600 mg for the celecoxib. No other adverse events or effects were observed during the trial. All outcomes were analyzed by intervention to treat for the 99 patients without any imputation of data for the missing control group patient.

As seen in Table 2, cumulative results for the 6 days, pain, nausea, itching, and oxycodone use were significantly lower in the intervention group. Most patients used no oxycodone in the intervention group. The sparsity of data prevented estimation of the difference of medians for nausea and itching.

The relative treatment effects for reducing opioid use and pain are displayed at 12- and 24-hour intervals in Figures 3 and 4. Use of oxycodone was less for the intervention group for each of the first 5 days. Pain differences by day were not significant. Data were too sparse for nausea

TABLE 2
Cumulative (6-day total) Pain, Nausea, Itching, and Oxycodone Use VRS Scores^a

VRS Score (0-10)	Blister Pack (n = 50)	Control (n = 49) ^b	Difference of Medians (95% CI)	P
Pain	28 (14 to 35)	35 (28 to 41)	-7 (-13 to -2)	.009
Nausea	0 (0 to 0)	0 (0 to 6)	NA ^c	.006
Itching	0 (0 to 0)	0 (0 to 4)	NA ^c	<.001
Oxycodone use, 5-mg tablets	0 (0 to 2)	8 (1.25 to 16)	-7 (-12 to -4)	<.001

^aResults are reported as median (interquartile range) unless otherwise indicated. Boldface *P* values indicate a statistically significant difference between study groups (*P* < .05). NA, not available; VRS, verbal rating scale.

^bOne control patient was lost to follow-up.

^cThe sparseness of data for nausea and itching prevented estimation of the difference of medians.

and itching to permit estimation of treatment effects across days.

DISCUSSION

In this prospective randomized clinical trial, the broad opioid-sparing effect produced by scheduled, simultaneous dosing of acetaminophen, celecoxib, and pregabalin reduced oxycodone requirements to zero for the majority of patients in the intervention group. The median (interquartile range) for total oxycodone tablets used by patients in the intervention group was 0 (0-2). Based on this study, we estimate that 90% of patients would require fewer than 8 oxycodone tablets during recovery. By contrast, 30 tablets are estimated to be required to achieve the same effect for patients in the control group. The reduction of opioid use observed in the study group was accompanied by significantly less overall pain.

After surgery, clinicians must balance the need to provide adequate analgesia in some patients with the concern of overprescribing opioids in others. Without a clinically validated alternative, surgeons are understandably reluctant to decrease opioid prescriptions in the setting of moderate to severe postoperative pain. This tendency to “overprescribe” is evidenced by estimates that 42% to 71% of prescribed opioid tablets remain unused.²

The ability for providers to confidently prescribe fewer opioids is paramount since the quantity received for a patient’s initial prescription is a major predictor of chronic opioid use.^{25,26} However, most existing “opioid-free” protocols simply restrict or eliminate access to opioids during surgical recovery while providing a variety of nonopioid alternatives.^{5,12,15,17,23} Thus, an important aspect of our trial was to provide nonopioid alternatives for patients as well as unrestricted access to opioids.

In this study, a prefilled blister pack system (Figure 1) was used to facilitate medication compliance and achieve simultaneous actions of pregabalin, acetaminophen, and celecoxib. This simplified packaging provided all 3 medications in 1 compartment to be taken at 6-hour intervals. Since the half-lives of celecoxib⁷ and pregabalin³ vary considerably from that of acetaminophen,¹⁰ their interval doses were adjusted accordingly. This novel form of drug delivery has important implications since each of these drugs has a unique opioid-sparing quality; however, their

simultaneous actions are rarely reported in an ambulatory setting.¹⁷

The analgesic regimen for patients in the control group was chosen to reflect current best practice protocols for pain control after ambulatory surgery.¹³ These protocols utilize peripheral nerve blocks, acetaminophen, and NSAIDs to optimize analgesia and minimize opioid requirements. Despite this extensive regimen, opioid requirements in the control group were significantly higher and more variable than for patients in the study group.

The prefilled medication blister pack featured in this trial (Figure 1) provided simultaneous effects of 3 analgesics with unique mechanisms of action. A secondary outcome measure for this study was opioid use after ambulatory center discharge. Existing calculations for “opioid or morphine equivalents” represent the best estimates available but are imperfect.¹⁹ To avoid this potential inaccuracy, a single opioid analgesic (oxycodone) was used for all patients after hospital discharge.

The nonopioid medications used for this trial are well tolerated after oral administration with minimal adverse effects. Each drug provides onset of analgesia within 30 to 40 minutes of oral administration. Finally, they are available in a cost-effective generic formulation. Techniques described for hospitalized patients such as parenteral ketamine, lidocaine infusions, and dexmedetomidine are not suitable for use after hospital discharge.⁸

Acetaminophen is an important component of multimodal analgesia with a mechanism of action that is unique from that of NSAIDs.¹⁰ Simultaneous administration of acetaminophen and NSAIDs produces improved analgesia and greater reductions in opioid requirements than either drug used in isolation.²¹

Celecoxib was chosen for this protocol because its selective COX-2 activity preserves platelet function and minimizes gastrointestinal side effects in cases requiring repeated dosing.⁷ A 400 mg dose of celecoxib has been shown in earlier studies to provide significant reduction in pain and postoperative opioid requirements.²⁴ Although it is commonly avoided in patients with an allergy to sulfonamide antimicrobials, available data do not support cross-reactivity between sulfonamide antimicrobials and other sulfonamide medications such as celecoxib.¹⁴

Pregabalin was included as the third component of multimodal analgesia for the study group. Gabapentinoids are known to reduce opioid requirements as well as

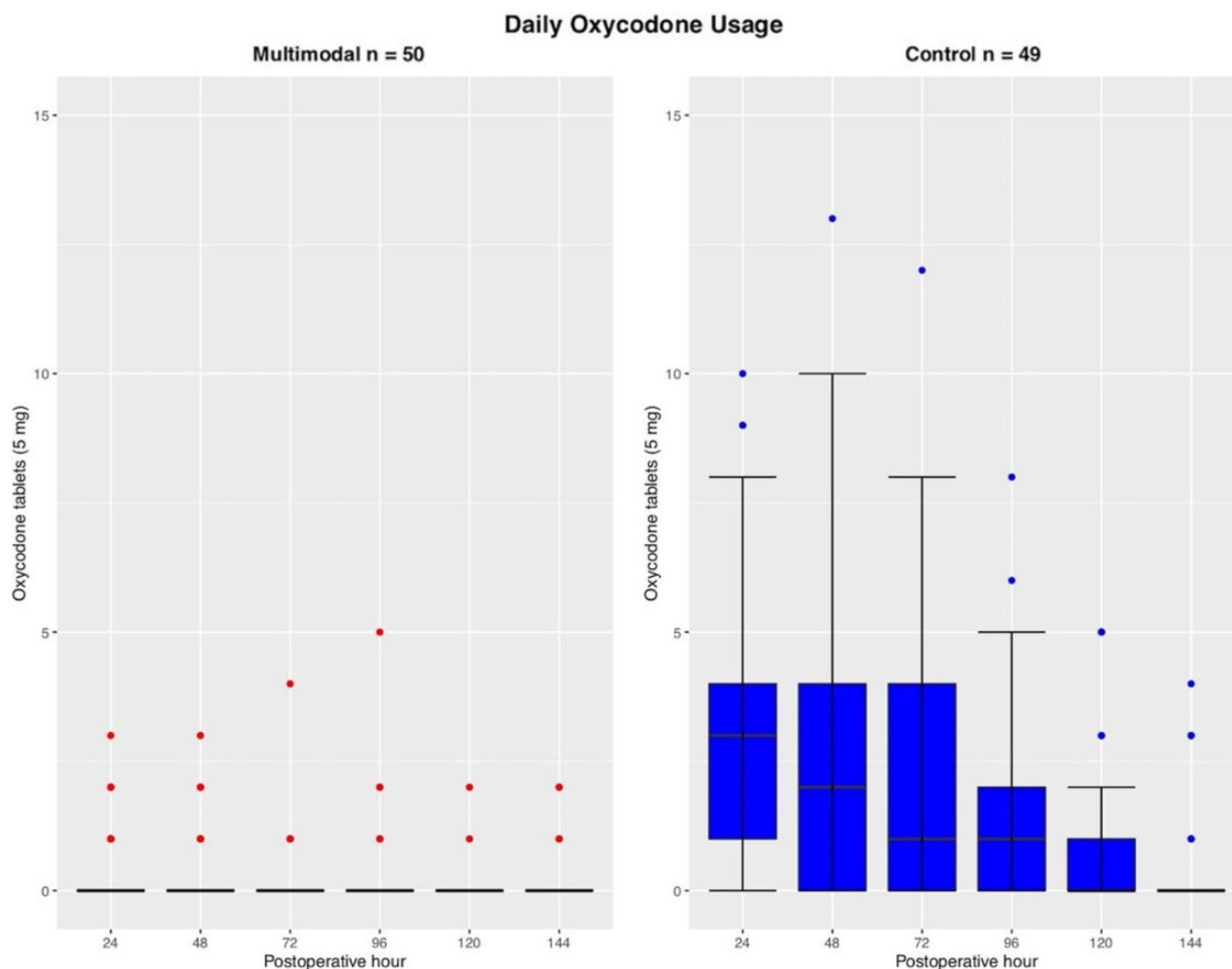


Figure 3. The counts of oxycodone tablet usage for each postoperative day are displayed by box plots. Each box goes from the 25th to the 75th percentile (interquartile range [IQR]) with a line indicating the median of the data. The whiskers start from the edge of the box and extend to the farthest data point that is within 1.5 times the upper bound of the IQR. The dots are the outlier values above the whiskers. No boxes are visible for the blister pack group, as the median, IQR, and whiskers were uniformly zero on all days. For oxycodone use, there was a difference ($F_{(1, \text{Inf})} = 50$; $P < .001$), a decrease over the 6 days ($F_{(3.7, \text{Inf})} = 32$; $P < .001$), and an interaction effect across days ($F_{(3.7, \text{Inf})} = 14$; $P < .001$) favoring the blister pack group. Benjamini-Yekutieli-adjusted P values for pairwise comparisons demonstrated that the oxycodone count for the blister pack group was less than that for the control group for days 1 to 5.

opioid-related side effects after surgery.^{16,28} Although both gabapentin and pregabalin have been used for acute pain management, pregabalin provides more predictable plasma concentrations and significantly faster onset of analgesia after oral administration.³ Side effects and adverse events associated with pregabalin are rare. In a recent meta-analysis of 281 clinical trials, serious adverse events such as respiratory failure, delirium, falls, and ataxia did not significantly increase with the use of gabapentinoids.²⁸

An important consideration when prescribing gabapentinoids is the potential for addiction or abuse. Recent reviews of pregabalin suggest that the risk of abuse in patients without a previous history of substance abuse disorder is very low.¹¹ The increased dopamine in the mesolimbic reward system produced by other drugs of abuse is not observed with pregabalin.⁴ However, for patients with

a history of substance abuse, gabapentinoids should be prescribed with caution and careful monitoring.

Respiratory depression is the primary cause of mortality associated with opioid analgesics.⁶ As such, the ventilatory effects produced by pregabalin alone and in combination with opioids are critical to define. In volunteers receiving pregabalin (150 mg) alone, there were no significant changes in minute ventilation, respiratory rate, or end-tidal CO_2 compared with placebo.¹⁸ For volunteers receiving target-controlled infusions of remifentanyl, the addition of pregabalin did not alter minute ventilation or respiratory rate but did cause a modest increase in end-tidal CO_2 (6 mm Hg) compared with remifentanyl alone.¹⁸

An important aspect of this study is the equitable treatment of pain for patients in the control group. Every

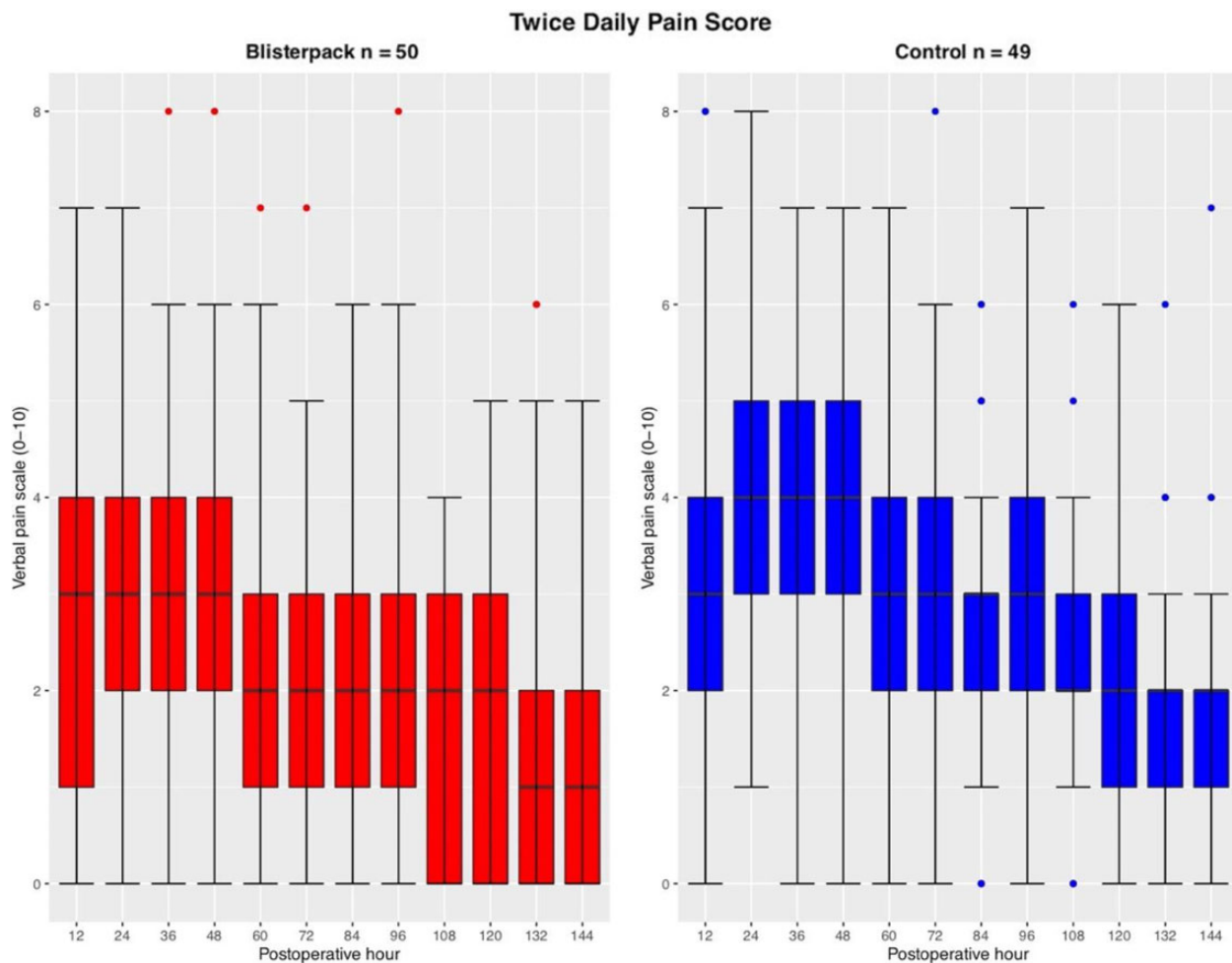


Figure 4. The verbal pain scores for each postoperative day are displayed by box plots. Each box goes from the 25th to the 75th percentile (interquartile range [IQR]), with a line indicating the median of the data. The whiskers start from the edge of the box and extend to the farthest data point that is within 1.5 times the upper bound of the IQR. The dots are the outlier values above the whiskers. There was a group difference ($F_{(1, \text{Inf})} = 7.5$; $P = .006$) and a decrease over the 6 days ($F_{(5.0, \text{Inf})} = 42$; $P < .001$) favoring the blister pack group, but no significant interaction effect across days ($F_{(5.0, \text{Inf})} = 1.4$; $P = .2$). There were no significant pairwise differences during the 6 days according to Benjamini-Yekutieli-adjusted P values.

consideration was made to ensure that pain control for the control group reflected existing best practice. To that end, patients in the control group received the same preoperative multimodal analgesics as those in the study group. Likewise, intraoperative anesthetics and postoperative nerve blocks were standardized between the 2 groups. Both groups received NSAIDs, oxycodone, and acetaminophen after hospital discharge.

For ethical reasons, a placebo-filled blister pack was not used, as this would not have permitted use of NSAIDs in patients in the control group. Oxycodone was combined with acetaminophen in patients in the control group to ensure that they benefited from the increased analgesia produced by combined acetaminophen and oxycodone compared with plain oxycodone.⁹ It should be noted, however, that patients in the control group were not prohibited from taking plain acetaminophen.

An important limitation of this trial is the inability to discern the true effects of medication delivery (blister pack vs no blister pack) and medications received (pregabalin vs no pregabalin). This effect could only be measured by studying these 2 interventions separately. It should also be noted that this blister pack system has been studied in a single procedure and the results should not be generalized without further clinical investigation. Similarly, this dosing regimen was applied to patients with a fixed range of weights and ages. There remains the opportunity to optimize dose response based on age and body mass.

A potential criticism of the enrollment process is that the study design may have discouraged patients who prefer opioid analgesia. This could cause a sampling bias leading only patients motivated to use less opioid to participate. However, only 2 eligible patients declined to participate after learning the objectives of this trial.

CONCLUSION

The blister pack system is novel as it provides scheduled, simultaneous action of 3 separate nonopioid analgesics. Its use is associated with reductions in both pain and opioid use after ACLR. The median (interquartile range) for total oxycodone tablets used by patients in the study group was 0 (0-2). Based on the results for the study group, it is predicted that a population of similar patients will have a 90% upper tolerance limit for opioid use during recovery of 8 oxycodone (5 mg) tablets. These findings allow practitioners to confidently prescribe lower amounts of opioid without concerns of undertreating pain when using this protocol. Further work on dosage of components within the blister pack should be performed to optimize the efficacy of nonopioid management of postoperative pain.

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